

AIM:

To write a program to find the solution of a matrix using Gaussian Elimination.

Equipments Required:

1. Hardware – PCs
2. Anaconda – Python 3.7 Installation / Moodle-Code Runner

Algorithm

1. Start the program
2. Import the necessary libraries(numpy, scipy.linalg)
3. Define the matrix using numpy
4. Use lu(), lu_solve(), lu_factor() to get the solutions
5. End the program

Program:

Program to find the solution of a matrix using Gaussian Elimination.

Developed by: vaishnavi s

RegisterNumber:212225230289

```
import os
os.environ["OPENBLAS_NUM_THREADS"]="1"

import numpy as np
import sys
n = int(input())
a = np.zeros((n,n+1))
x = np.zeros(n)
for i in range(n):
    for j in range(n+1):
        a[i][j] = float(input())
for i in range(n):
    if a[i][i] == 0.0:
        sys.exit('Divide by zero detected!')
    for j in range(i+1, n):
        ratio = a[j][i]/a[i][i]
        for k in range(n+1):
            a[j][k] = a[j][k] - ratio * a[i][k]
```

```

x[n-1] = a[n-1][n]/a[n-1][n-1]
for i in range(n-2,-1,-1):
    x[i] = a[i][n]
    for j in range(i+1,n):
        x[i] = x[i] - a[i][j]*x[j]
    x[i] = x[i]/a[i][i]

for i in range(n):
    print('X%d = %0.2f' %(i,x[i]), end = ' ')

```

Output:

The screenshot shows a web browser with a Python code editor. The code implements Gaussian elimination. The output shows the solution for a system of linear equations.

```

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4 ...
5 import os
6 os.environ["OPENBLAS_NUM_THREADS"]="1"
7
8 import numpy as np
9 import sys
10 n = int(input())
11 a = np.zeros((n,n+1))
12 x = np.zeros(n)
13 for i in range(n):
14     for j in range(n+1):
15         a[i][j] = float(input())
16 for i in range(n):
17     if a[i][i] == 0.0:
18         sys.exit('Divide by zero detected!')
19     for j in range(i+1, n):
20         ratio = a[j][i]/a[i][i]
21         for k in range(n+1):
22             a[j][k] = a[j][k] - ratio * a[i][k]
23 x[n-1] = a[n-1][n]/a[n-1][n-1]
24 for i in range(n-2,-1,-1):
25     x[i] = a[i][n]
26     for j in range(i+1,n):
27         x[i] = x[i] - a[i][j]*x[j]
28     x[i] = x[i]/a[i][i]
29
30 for i in range(n):
31     print('X%d = %0.2f' %(i,x[i]), end = ' ')

```

The output shows the solution for a system of linear equations:

Input	Expected	Got
3	X0 = 53.35 X1 = -8.88 X2 = 4.40	X0 = 53.35 X1 = -8.88 X2 = 4.40

Result:

Thus the program to find the solution of a matrix using Gaussian Elimination is written and verified using python programming.